THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — MATHS FULL SYLLABUS PAPER SET 07

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The reflection of the point $(5,\ 7)$ in the y-axis is: 1

(a) $(-5,\ -7)$ **(b)** $(5,\ -7)$ **(c)** $(7,\ 5)$ **(d)** $(-5,\ 7)$

2. The sequence $3, 7, 11, 15, \ldots$ forms a linear pattern. Its nth term is: 1

(a) $4n - 1$ **(b)** $3n + 4$ **(c)** $4n - 7$ **(d)** $4n + 3$

3. The decimal expansion of $\frac{13}{25}$ is: 1

(a) non-terminating repeating **(b)** non-terminating non-repeating

(c) terminating **(d)** cannot be determined

4. The expression $49x^2 + 28xy + 4y^2$ can be factorised as: 1

(a) $(7x + 4y)^2$ **(b)** $(49x + 2y)^2$ **(c)** $(7x - 2y)^2$ **(d)** $(7x + 2y)^2$

5. The diameter of a circle is 26 cm and a chord is 24 cm long. The distance of the chord from the centre is: 1

(a) 1 cm **(b)** 5 cm **(c)** 13 cm **(d)** 25 cm

6. The length of an arc subtending 180° at the centre of a circle of radius 14 cm ($\pi = \frac{22}{7}$) is: 1

(a) 44 cm **(b)** 22 cm **(c)** 46 cm **(d)** 88 cm

7. The game of Snakes and Ladders originated in ancient India from a game called: 1

(a) Jñān-Chaupaḍ **(b)** Mancala **(c)** Pachisi only **(d)** Shatranj

8. The sum of the first 50 natural numbers is: 1

(a) 1274 **(b)** 1275 **(c)** 2550 **(d)** 1276

9. A proposition is a statement that is: 1

(a) a question **(b)** always false

(c) either true or false **(d)** always true

10. "If $a = b$, then $a + 5 = b + 5$." This proposition and its converse are: 1

(a) both true **(b)** both false

(c) only the proposition is true **(d)** only the converse is true

11. How much water must be added to 12 litres of a 10% solution to reduce its concentration to 8%? 1

(a) 4 litres **(b)** 5 litres **(c)** 6 litres **(d)** 3 litres

12. The average of 12 values is 85. When one more value, 33, is added, the new average is: 1

(a) 82 **(b)** 83 **(c)** 81 **(d)** 80

13. While adding 268 and 175 by the column algorithm, the number of columns that produce a carry of 1 is: 1

(a) 1 **(b)** 2 **(c)** 3 **(d)** 4

14. The improved gcd algorithm that keeps most-recent-common-divisor scans k from 2 up to min(12, 30). The number of values of k scanned is: 1

(a) 12 **(b)** 11 **(c)** 29 **(d)** 10

15. **Assertion (A):** The centroid of a triangle divides every median in the ratio 2 : 1. 1
Reason (R): The medians of a triangle are concurrent.

(a) Both A and R are true and R is the correct explanation of A

(b) Both A and R are true but R is not the correct explanation of A

(c) A is true but R is false

(d) A is false but R is true

16. The sides of a triangle are 12 cm, 16 cm and 20 cm. The perimeter of the triangle formed by joining the midpoints of its sides is: 1

(a) 46 cm **(b)** 26 cm **(c)** 48 cm **(d)** 24 cm

17. For which value of k does the pair kx + 2y = 5 and 3x + 6y = 12 have no solution? 1

(a) 5 **(b)** 1 **(c)** 0 **(d)** 3

18. Taking $\pi = \frac{22}{7}$, the surface area of a sphere of radius 7 cm is: 1

(a) 154 cm^2 **(b)** 1232 cm^2 **(c)** 616 cm^2 **(d)** 308 cm^2

19. The total surface area of a cuboid of length 10 cm, width 6 cm and height 4 cm is: 1

(a) 124 cm^2 **(b)** 240 cm^2 **(c)** 134 cm^2 **(d)** 248 cm^2

20. **Assertion (A):** If the radius of a sphere is doubled, its surface area is also doubled. 1
Reason (R): The surface area of a sphere is $4\pi r^2$.

(a) Both A and R are true and R is the correct explanation of A

(b) Both A and R are true but R is not the correct explanation of A

(c) A is true but R is false

(d) A is false but R is true

SECTION — B (Very Short Answer Type) 8 × 2 = 16

Answer in about 20–30 words each.

21. Write the coordinates of the vertices of a rectangle with one vertex at the origin, length 7 units along the x-axis and breadth 4 units along the y-axis (in Quadrant I). 2

22. Among all rectangles with perimeter 40 units, which has the greatest area? State that area. 2

23. One month of the year is chosen at random. What is the probability that it has 31 days? 2

24. A taxi charges a fixed ₹200 and ₹40 per km. Write the fare after 1, 2, 3 km and find the fare for 10 km. 2

25. Write the converse of "If n is the square of a prime, then n has exactly 3 factors." Give a counterexample if it is false. 2

26. In a stacked bar chart, what does the total length of a bar represent? 2

27. Can a quadrilateral have angles 100°, 80°, 120° and 70°? Give a reason. 2

28. The point $(1, k)$ lies on $3x + 2y = 7$. Find k. 2

SECTION — C (Short Answer Type) 8 × 3 = 24

Answer in about 40–60 words each.

29. A spinner has 8 equal sectors numbered 1 to 8. Find the probability that it stops on 3
(i) a prime
(ii) a number greater than 5.

30. Describe an algorithm to find the least common multiple (lcm) of two numbers m and n. 3

31. The areas of three faces of a cuboid that meet at a corner are 12 cm^2, 20 cm^2 and 15 cm^2. Find its volume. 3

32. The balance of a prepaid plan is ₹600 and it reduces by ₹15 daily. Write $b(x)$ and find after how many days it runs out. 3

33. A ball dropped from 24 ft rebounds to $\frac{3}{4}$ of its previous height each time. Find the heights after the first three bounces and state the type of sequence. 3

34. The breadth of a pool is 4 m less than its length and its area is 96 m². Find its dimensions. 3

35. Use $F = \frac{9}{5}C + 32$ to find 3
(i) the Celsius value corresponding to 212 °F,
(ii) the temperature at which Celsius and Fahrenheit readings are equal.

OR

Write $y = \frac{2}{3}x - 4$ in the form $ax + by + c = 0$ with integer coefficients and verify that $(6, 0)$ and $(3, -2)$ lie on it. 3

36. Write the converse of “If a triangle is isosceles, then it has two equal angles”. Is it true? Briefly justify using an altitude. 3

OR

If n and $n + 3$ have no common factor, then n is not a multiple of 3. Write the converse. Is the converse true? 3

SECTION — D (Long Answer Type) 4 × 5 = 20

Answer in about 100–150 words each.

37. **Case study:** A teacher gives two options for the final grade: Option 1 uses the plain average of project (80), test (60) and exam (70). Option 2 uses weights 1 : 2 : 5. 5
(i) Find both.
(ii) Which option favours a student who is strong in the exam?
(iii) Name the average used in Option 2.

OR

(i) Make a 100% stacked bar table for the generation data (MW): 2015 — coal 600, solar 100, wind 200, hydro 100; 2025 — coal 800, solar 600, wind 400, hydro 200. 5
(ii) What do you observe about solar?

38. (i) Test whether the triples 5
(a) 15, 20, 25 and
(b) 11, 60, 62 can form right-angled triangles.
(ii) Explain why a triple that satisfies $a^2 + b^2 = c^2$ must form a right-angled triangle.

OR

Case study: Meera notices: “All the rectangles I drew had equal diagonals.” 5

(i) Write this as a proposition.

(ii) Write the converse.

(iii) Draw (describe) an isosceles trapezium to show the converse is false.

(iv) Name a type of quadrilateral for which the converse is true under an extra condition.

39. Find the point on the x-axis which is equidistant from $(-2,\ 5)$ and $(2,\ -3)$, and find the common distance. 5

OR

Find the value(s) of k if the distance between $(k,\ 2)$ and $(3,\ -4)$ is 10 units, and write the corresponding points. 5

40. **Case study:** Two players A and B play with a spinner of 8 equal sectors numbered 1–8. A wins if the number is prime; B wins if the number is a composite number. 5

(i) Find $P(A \text{ wins})$.

(ii) Find $P(B \text{ wins})$.

(iii) Is the game fair?

(iv) Which number(s) make neither win?

OR

A card is drawn from a pack of 52. Let E be the event ‘the card is a spade’. 5

(i) Find $P(E)$.

(ii) Find $P(\text{not } E)$ and show it equals $1 - P(E)$.

(iii) Find the probability of ‘a spade or a heart’.

(iv) Find the probability of ‘an ace or a king’.

END OF QUESTION PAPER **Total: 80 Marks**

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